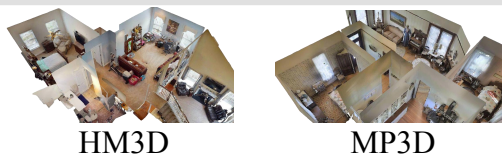
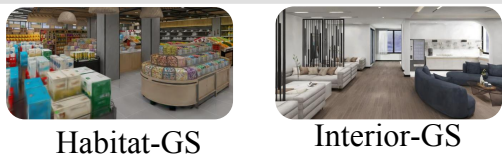


## Diverse Envolving Worlds

### A Standard Simulation Worlds



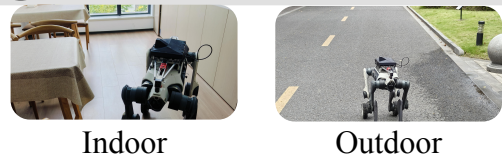
### B Photorealistic Simulation



### C Evolving Simulation Worlds



### D Real-World Evolution



Memory  
Construction

#### 1 Observed History



#### 2 Hidden Evolution

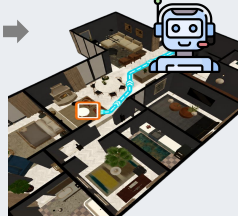


#### 3 Current Belief



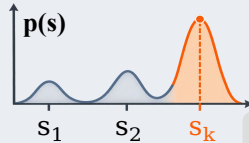
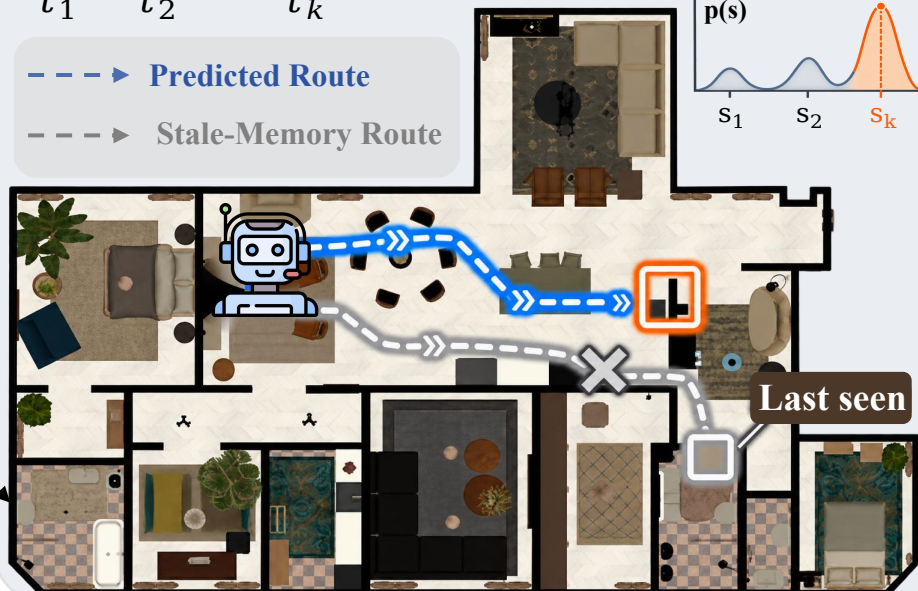
Memory  
Update

#### 4 Belief-guided Plan/Act



---> Predicted Route

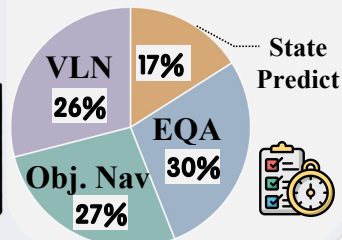
---> Stale-Memory Route



## P4D-Bench

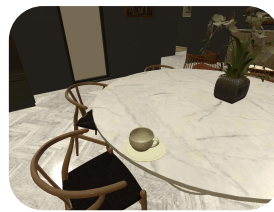


54 Scenes  
803.68k Tasks  
4860 Scene-Days  
239.19k Dynamic  
State Changes



## Unified Embodied Tasks

### A Embodied QA



Q: Where did I put the cup after I left the dining table ?

### B Active Exploration



### C Predictive Navigation



### D Real World Interaction



Task: Find the lounge on this floor and enter it.